Mathematical Equations of Memory Systems

Chapter 7: Quantitative Models of Memory Dynamics and Consolidation

Tanveer Hussain

April 2, 2026

Abstract

This document presents mathematical models for memory systems across all timescales — from working memory dynamics to long-term consolidation. Models cover persistent firing and oscillatory mechanisms in working memory, synaptic plasticity (LTP/LTD), memory consolidation, systems-level transfer from hippocampus to cortex, sleep-dependent consolidation dynamics, and spatial cognition. These equations provide computational frameworks for understanding memory formation, storage, retrieval, and failure in aging and disease.

Contents

1 Working Memory Dynamics **2**
- 1.1 Persistent Firing: PFC Delay Activity Maintenance . . . 2
- 1.2 Theta-Gamma Oscillatory Coupling: Item Segregation . . . 2

2 Synaptic Plasticity **3**
- 2.1 NMDA-Dependent LTP: Calcium Influx and CaMKII Activation . . . 3
- 2.2 CaMKII Autophosphorylation Switch Model . . . 3
- 2.3 AMPA Receptor Trafficking: Exocytosis and Endocytosis . . . 3
- 2.4 Spike-Timing Dependent Plasticity (STDP): Temporal Window . . . 4

3 Memory Consolidation **4**
- 3.1 Systems Consolidation: Hippocampus-to-Cortex Transfer . . . 4
- 3.2 Sleep-Dependent Consolidation: Sharp-Wave Ripple Replay . . . 5
- 3.3 Three-Way Coupling: Slow Oscillations, Spindles, and Ripples . . . 5

4 Memory Retrieval and Reconsolidation **5**
- 4.1 Pattern Completion: Cue-Driven Retrieval . . . 5
- 4.2 Reconsolidation Dynamics: Labilization and Restabilization . . . 6

5 Spatial Cognition **6**
- 5.1 Place Field Firing: Gaussian Profile . . . 6
- 5.2 Grid Cell Hexagonal Firing: Regular Spacing . . . 7

6 Case Studies **7**
6.1 Case Study 1: Persistent Firing in a PFC Network 7
6.2 Case Study 2: LTP Induction and CaMKII Activation 7
6.3 Case Study 3: Consolidation Timeline 7
6.4 Case Study 4: Sleep Spindle Nesting 7
6.5 Case Study 5: Place Cell Remapping 8

1 Working Memory Dynamics

1.1 Persistent Firing: PFC Delay Activity Maintenance

Working memory maintains information during delays via elevated firing rates in prefrontal cortex (PFC). Recurrent NMDA-mediated excitation sustains activity through a bistable network state.

$$\frac{dV}{dt} = -V + W_{\text{rec}} f(V) + I_{\text{ext}} + \eta(t) \tag{1}$$

where $f(V) = \frac{V^2}{V_0^2 + V^2}$ (squared nonlinearity models NMDA gating).

Variable	Definition	Units	Typical Value
V	Population firing rate (PFC neurons)	spike/s	5–50
W_{rec}	Recurrent NMDA-mediated weight	dimensionless	1.1–1.5
I_{ext}	External input (stimulus or cue)	mV	0–10
V_0	NMDA half-activation voltage	mV	5
$\eta(t)$	Noise	spike/s	—

Explanation: At stimulus onset, I_{ext} drives firing. When stimulus ends, recurrent weight $W_{\text{rec}} > 1$ sustains elevated activity via positive feedback, maintaining a "bump" attractor state. Removal of working memory item requires lateral inhibition or top-down suppression.
Key Reference: Wang, X. J. (2001). Synaptic reverberation underlying mnemonic persistent activity. *Trends in Neurosciences*, 24(8), 455–463.

1.2 Theta-Gamma Oscillatory Coupling: Item Segregation

Working memory items are segregated across theta phases ($\sim$ 4-8 Hz), with each item encoded in gamma bursts ($\sim$ 30-80 Hz) nested within a theta cycle.

$$\text{MI}(f_1, f_2) = \frac{\sum_n P(f_1^n) \log\left(\frac{P(f_1^n, f_2)}{P(f_1^n)P(f_2)}\right)}{\log(N)} \tag{2}$$

Variable	Definition
MI	Modulation Index (0–1; higher = stronger coupling)
f_1, f_2	Frequency bands (e.g., theta and gamma)
$P(f_1^n)$	Phase bin probability
$P(f_1^n, f_2)$	Joint probability
N	Number of phase bins

Explanation: Measures coupling between two frequency bands. Higher MI indicates that gamma power is more tightly modulated by theta phase, improving information segregation.
Key Reference: Lisman, J. E., & Jensen, O. (2013). The theta-gamma neural code. *Neuron*, 77(6), 1002–1016.

---

2 Synaptic Plasticity

2.1 NMDA-Dependent LTP: Calcium Influx and CaMKII Activation

$$\frac{d[\mathrm{Ca}^{2+}]_i}{dt} = g_{\mathrm{NMDA}}(V_m - E_{\mathrm{Ca}}) \cdot B_{\mathrm{Mg}}(V_m) - k_{\mathrm{pump}}[\mathrm{Ca}^{2+}]_i \tag{3}$$

Variable	Definition	Typical Value
g_{NMDA}	NMDA conductance (mV^{-1})	0.1–1.0
V_m	Membrane voltage	Variable
E_{Ca}	Calcium reversal potential	+130 mV
B_{Mg}	Mg^{2+} blockade factor	0–1
k_{pump}	Calcium extrusion rate	0.01–0.1 ms^{-1}

Explanation: Ca^{2+} influx through NMDA receptors is the trigger for LTP induction. Voltage-dependent Mg^{2+} block is relieved by depolarization, allowing Ca^{2+} entry. High-frequency stimulation causes large, sustained Ca^{2+} rise triggering LTP; low-frequency stimulation causes smaller Ca^{2+} rise triggering LTD.
Key Reference: Jahr, C. E., & Stevens, C. F. (1990). Voltage dependence of NMDA-activated conductances. *Journal of Neuroscience*, 10(9), 3178–3182.

2.2 CaMKII Autophosphorylation Switch Model

$$\frac{d[\mathrm{pCaMKII}]}{dt} = k_{\mathrm{on}}[\mathrm{Ca}^{2+}][CaMKII] - k_{\mathrm{off}}[\mathrm{pCaMKII}] \tag{4}$$

Variable	Definition	Typical Value
[pCaMKII]	Phosphorylated CaMKII concentration	μM
k_{on}	Activation rate constant	0.001–0.1 $\mu\mathrm{M}^{-1}\ \mathrm{s}^{-1}$
k_{off}	Deactivation rate constant	0.0001–0.01 s^{-1}

Explanation: CaMKII autophosphorylation at Thr286 creates a phosphatase-resistant form that persists even after Ca^{2+} returns to baseline. This serves as a molecular "memory" switch lasting hours to days.
Key Reference: Lisman, J., Schulman, H., & Cormier, R. H. (2002). Molecular mechanisms of memory. *Nature Reviews Neuroscience*, 3(3), 175–190.

2.3 AMPA Receptor Trafficking: Exocytosis and Endocytosis

$$\frac{d[R_{\mathrm{surf}}]}{dt} = k_{\mathrm{in}}[\mathrm{pCaMKII}][R_{\mathrm{pool}}] - k_{\mathrm{out}}[R_{\mathrm{surf}}] - k_{\mathrm{deg}}[R_{\mathrm{surf}}] \tag{5}$$

Variable	Definition
$[R_{\text{surf}}]$	Surface AMPA receptors (number per synapse)
$[R_{\text{pool}}]$	Intracellular AMPA receptor pool
k_{in}	Exocytosis rate (pCaMKII-dependent)
k_{out}	Endocytosis rate
k_{deg}	Baseline degradation rate

Explanation: During early-phase LTP, pCaMKII drives exocytosis of AMPA receptors from recycling endosomes, inserting them into the postsynaptic membrane. This increases synaptic strength within minutes.

2.4 Spike-Timing Dependent Plasticity (STDP): Temporal Window

$$\Delta w = \begin{cases} A_+ e^{-|\Delta t|/\tau_+} & \text{if } \Delta t > 0 \text{ (LTP)} \\ -A_- e^{-|\Delta t|/\tau_-} & \text{if } \Delta t < 0 \text{ (LTD)} \end{cases} \tag{6}$$

Parameter	Definition	Typical Value
Δt	Time difference (pre – post)	$\pm$ 100 ms
τ_+	LTP time constant	16.8 ms
τ_-	LTD time constant	33.7 ms
A_+	LTP amplitude	0.005–0.05
A_-	LTD amplitude	0.003–0.03

Explanation: Synaptic weight changes depend on precise timing of pre- and postsynaptic spikes. Pre-before-post (causal) leads to LTP; post-before-pre (anti-causal) leads to LTD. Asymmetry with $\tau_- > \tau_+$ and $A_+ > A_-$ stabilizes learning.
Key Reference: Song, S., Miller, K. D., & Abbott, L. F. (2000). Competitive Hebbian learning through spike-timing-dependent synaptic plasticity. *Nature Neuroscience*, 3(9), 919–926.

---

3 Memory Consolidation

3.1 Systems Consolidation: Hippocampus-to-Cortex Transfer

$$\frac{dM_{\text{ctx}}}{dt} = \alpha M_{\text{hip}} - \beta M_{\text{ctx}} \tag{7}$$

Variable	Definition	Typical Value
M_{hip}	Hippocampal memory strength	Arbitrary units
M_{ctx}	Cortical memory strength	Arbitrary units
α	Transfer rate from hippocampus	0.01–0.1 day^{-1}
β	Cortical decay/forgetting rate	0.001–0.01 day^{-1}

Explanation: Models gradual consolidation over days to weeks. Hippocampus initially stores strong memory trace (M_{hip} high); with time, cortex accumulates memory strength while hippocampus trace decays (though traces may remain per MTT).
Key Reference: Squire, L. R., & Alvarez, P. (1995). Retrograde amnesia and memory consolidation: A neurobiological perspective. *Current Opinion in Neurobiology*, 5(2), 169–177.

3.2 Sleep-Dependent Consolidation: Sharp-Wave Ripple Replay

$$\text{Replay fidelity} = \frac{\text{Cov}(\text{awake spike sequence}, \text{ripple spike sequence})}{\sigma_{\text{awake}}\sigma_{\text{ripple}}} \tag{8}$$

Explanation: Pearson correlation between population firing patterns during awake experience and subsequent SWR replay. Values: 0.5–0.9 (forward replay), –0.3–0.6 (reverse replay). Higher fidelity predicts better memory retention.
Key Reference: Wilson, M. A., & Louie, B. (2001). Temporally structured replay of awake hippocampal ensemble activity during rapid eye movement sleep. *Neuron*, 29(1), 145–156.

3.3 Three-Way Coupling: Slow Oscillations, Spindles, and Ripples

$$\phi_{\text{spindle}}(t) \subset \phi_{\text{SO up-state}}(t) \subset \phi_{\text{SWR}}(t) \tag{9}$$

Explanation: Thalamocortical sleep spindles (12–16 Hz) occur within the up-states of cortical slow oscillations (~ 0.75 Hz), which are coupled with hippocampal sharp-wave ripples (100–250 Hz). This hierarchical coordination provides precise timing for transfer of information from hippocampus to cortex during sleep.
Key Reference: Sirota, A., Csicáná, T., Buzsáki, G., & Buhl, D. L. (2003). Communication between neocortex and hippocampus during sleep in rodents. *Proceedings of the National Academy of Sciences*, 100(4), 2065–2069.

---

4 Memory Retrieval and Reconsolidation

4.1 Pattern Completion: Cue-Driven Retrieval

$$\mathbf{r}^{\text{retrieved}}(t) \to \mathbf{r}^{\text{full}} \quad \text{via} \quad \mathbf{r}(t+1) = \text{sign}(W\mathbf{r}(t) + \text{cue}) \tag{10}$$

Variable	Definition
$\mathbf{r}^{\text{retrieved}}$	Retrieved population firing pattern from partial cue
$\mathbf{r}^{\text{full}}$	Complete stored memory pattern
W	Recurrent connection matrix (Hebbian)

Explanation: Hippocampal CA3 recurrent networks enable retrieval of full memories from partial cues through iterative dynamics. Cue activates neurons; recurrent connectivity drives convergence to stored attractor (complete memory). This is the neural basis of pattern completion.
Key Reference: O'Reilly, R. C., & McClelland, J. L. (1994). Hippocampal conjunctive encoding, storage, and recall. *Hippocampus*, 4(6), 661–682.

4.2 Reconsolidation Dynamics: Labilization and Restabilization

$$\text{Stability}(t) = \begin{cases} S_{\text{high}} & 0 \leq t < t_{\text{retrieval}} \text{ (consolidated)} \\ S_{\text{low}} & t_{\text{retrieval}} \leq t < t_{\text{retrieval}} + t_w \text{ (labile window)} \\ S_{\text{high}} & t > t_{\text{retrieval}} + t_w \text{ (reconsolidated)} \end{cases} \tag{11}$$

Variable	Definition	Typical Value
t_w	Reconsolidation window	5–6 hours
$t_{\text{retrieval}}$	Time of memory reactivation	Event time

Explanation: Memory retrieval transiently destabilizes the trace through unknown mechanisms (possibly involving ubiquitin-proteasome degradation of PSD proteins or immediate early gene dysregulation). During the window t_w, new protein synthesis is required for restabilization. Disrupting synthesis (protein synthesis inhibitors, β-blockers) during this window weakens or erases the memory.
Key Reference: Nader, K., Schafe, G. E., & LeDoux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature*, 406(6797), 722–726.

---

5 Spatial Cognition

5.1 Place Field Firing: Gaussian Profile

$$r(\mathbf{x}) = r_{\max} \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_0\|^2}{2\sigma^2}\right) \tag{12}$$

Variable	Definition	Typical Value
$r(\mathbf{x})$	Firing rate at location $\mathbf{x}$	spike/s
$r_{\max}$	Peak firing rate in field	5–100 spike/s
$\mathbf{x}_0$	Center of place field	Environment-dependent
σ	Place field width	10–100 cm

Explanation: CA1 and CA3 place cells fire when animal occupies a specific region (place field). Firing is well-fit by 2D Gaussian centered at $\mathbf{x}_0$ with width σ. Place field size increases from dorsal (small, precise) to ventral hippocampus (large, coarse).
Key Reference: O'Keefe, J., & Burgess, N. (2005). Dual phase and rate coding in hippocampal place cells: Theoretical significance and relationship to entorhinal grid cells. *Hippocampus*, 15(7), 853–866.

5.2 Grid Cell Hexagonal Firing: Regular Spacing

$$\text{Grid spacing} = \lambda = \alpha \cdot d_{\text{AP}} \tag{13}$$

Variable	Definition	Typical Value
λ	Inter-peak distance in grid	30–300 cm
d_{AP}	Distance along anterior-posterior axis (EC)	—
α	Scaling factor	0.6–1.0

Explanation: Entorhinal grid cells exhibit hexagonal firing patterns with spacing that increases along the dorsal-to-ventral axis. This provides a multi-scale coordinate system: fine grids (dorsal) for precise location, coarse grids (ventral) for broader context.
Key Reference: Hafting, T., Fyhn, M., Molden, S., Moser, M. B., & Moser, E. I. (2005). Microstructure of a spatial map in the entorhinal cortex. *Nature*, 436(7052), 801–806.

---

6 Case Studies

6.1 Case Study 1: Persistent Firing in a PFC Network

A PFC network has $W_{\text{rec}} = 1.3$, $V_0 = 5$ mV, $\tau = 100$ ms. During stimulus ($0 < t < 500$ ms), $I_{\text{ext}} = 8$ mV. After stimulus, $I_{\text{ext}} = 0$.

Stable fixed point during stimulus: $V^* = 40$ spike/s. After stimulus removal, network maintains $V \approx 38$ spike/s for ~ 2 seconds due to recurrent weight > 1. Lateral inhibition or attention signals (top-down suppression) can reset activity to baseline.

6.2 Case Study 2: LTP Induction and CaMKII Activation

HFS (100 Hz, 100 pulses) produces peak $[\text{Ca}^{2+}] = 2.0$ μM lasting 50 ms. At such levels, CaMKII phosphorylation fraction reaches $\sim 80\%$. Autophosphorylation stabilizes $\sim 40\%$ of pCaMKII constitutively active (independent of Ca^{2+}) for 4–6 hours, supporting E-LTP to L-LTP transition.

6.3 Case Study 3: Consolidation Timeline

Day 0 (encoding): $M_{\text{hip}} = 100$, $M_{\text{ctx}} = 5$. With $\alpha = 0.05$ day^{-1}, $\beta = 0.005$ day^{-1}:

Day 7: $M_{\text{hip}} \approx 70$, $M_{\text{ctx}} \approx 35$. Day 30: $M_{\text{hip}} \approx 22$, $M_{\text{ctx}} \approx 80$.

Memory becomes hippocampus-independent by 30 days post-encoding, consistent with standard consolidation timeline.

6.4 Case Study 4: Sleep Spindle Nesting

Slow oscillation: 0.75 Hz (period 1333 ms); up-state lasts ~ 800 ms. Sleep spindles (12–16 Hz, here 14 Hz, period ~ 71 ms) nest within up-state: 11 spindle cycles per up-state. SWRs (150 Hz, period ~ 6.7 ms) are 12 ripples per spindle. This triple nesting creates precise temporal window for cortical synaptic potentiation.

6.5 Case Study 5: Place Cell Remapping

Recording same CA1 neuron in two environments:

Environment A: Place field centered at $(50\text{ cm}, 30\text{ cm})$, $\sigma = 20$ cm, $r_{\max} = 25$ spike/s.

Environment B (rotated 90°): Place field center rotates to $(30\text{ cm}, 50\text{ cm})$ (coordinate rotation), $\sigma = 20$ cm, $r_{\max} = 23$ spike/s.

Complete remapping of place field population indicates creation of a new hippocampal representation for the novel environment.

Next: Chapter 8 — Consciousness and Awareness